

Varada Kalyan

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Summary

M.Tech in **Artificial Intelligence** with a strong foundation in **software development, automation, and full-stack application design**. Experienced in **Python, Java, React Native, Flask, and SQL**, with hands-on exposure to AI concepts and system automation. Proven ability to design and deploy scalable applications, including **GNN based recommendation system, POS billing systems, and web-based platforms**. Skilled in API integration, database management, and GUI development. Strong **problem-solving and collaborative abilities**, with experience in both individual and team-driven projects. Seeking challenging roles in **AI development or software engineering** to leverage technical expertise and deliver innovative solutions.

Skills

Programming & Scripting: Java, SQL, Python, C++.

Libraries/Frameworks: NumPy, Pandas, PyTorch, Tkinter, PySimpleGUI, Paramiko, Selenium, Qt6.

Other Skills: Continuous Learning, Troubleshooting, Problem Solving, LLMs.

Education

Amrita Vishwa Vidyapeetham, M.Tech in Artificial Intelligence | CGPA 8.19/10 Aug 2024 - May 2026

- Machine Learning, Deep Learning, Mathematical Foundations of Computing

Sastra University, B.Tech in Electronics & Communication Engg | CGPA 7.22/10 Jun 2017 - May 2021

- Computer Networks, Digital Electronics

Experience

Test Engineer, Infosys Limited, Boeing client – Bangalore, IN Sept 2021 - Feb 2024

- Performed manual **QA testing** for web-based applications and aviation documentation (HTML, XML), ensuring compliance with Boeing standards.
- Reviewed and verified **confidential flight manufacturing specifications and wiring schematics**, conducting error-checks and reporting issues.
- **Automated repetitive testing tasks** (links, buttons, workflows) using **Selenium with Java and Python**, improving efficiency.
- **Collaborated with cross-functional teams** to ensure timely project delivery and accuracy in all documentation reviews.

Projects

Robot body generation using Language Model College Research Project
(Python, LLM, SLM, EvoGym, Soft Robotics)

- Finetuned a small language model (**Llama 3B**) using **PEFT** and **QLora** techniques in local system for soft robot body structure generation.
- Used available robot body structure **dataset from EvoGym** and also evolved a custom dataset using EvoGym framework.
- Planning to finetune for **robot bodies suitable for multiple environments** like Walker, Flipper etc using a custom dataset generated by **Genetic Algorithm, Proximal Policy Optimization**.

Custom Product Recommendation System using GNN

(Python, GNN, GraphSAGE, Custom dataset)

- Developed a product recommendation system using **Graph Neural Networks (GNN)** by modeling co-purchase relationships from **500+ real billing records** into a product-product graph.
- Designed and implemented a **GraphSAGE-based** neural architecture using **PyTorch Geometric** to learn product

embeddings and predict item similarity for personalized recommendations.

- Achieved efficient and interpretable recommendations by applying **embedding similarity (cosine distance)** and integrating the model output into a retail billing workflow for real-time suggestions.

Supermarket Billing Application

[Github Repository link](#)

(Python, Tkinter, SQLAlchemy)

- Built and deployed a custom billing and inventory management system using **Python, Tkinter, and SQLAlchemy on Linux OS**.
- Integrated barcode scanning and ESC/POS thermal printer to **automate checkout processes**, improving operational efficiency.
- **Configured Linux OS** and optimized system performance for seamless application functionality.
- Designed a **user-friendly GUI tailored to supermarket** workflows, reducing staff effort and training time.

Hospital Appointment Management Website

[Github Repository link](#)

(Flask, SQLAlchemy, HTML, Bootstrap)

- Developed a hospital appointment management website using **Flask and SQLAlchemy** for efficient database management.
- Implemented backend logic and optimized code to **handle patient records and appointment histories** seamlessly.
- Designed responsive web pages using **HTML and Bootstrap**, improving usability across devices.
- Ensured **secure storage and management of patient data**, supporting streamlined administrative operations.

E-Commerce Mobile App

[Github Repository link](#)

(React Native, JavaScript, Firebase, Springboot)

- Developed a customer-facing mobile app for **order tracking, payments, and product delivery** using React Native, expo and Firebase.
- Implemented **real-time database and Firebase Authentication** for secure user management.
- Integrated **geolocation APIs** to manage delivery zones and enhance logistics efficiency.
- Designed **server-driven UI updates** to enable **weekly feature releases** and increase user engagement.

Languages Known

English | Telugu | Tamil | Hindi | Kannada